

QUBIT 2026 HACKATHON · CHALLENGE: ANTENNA TILT

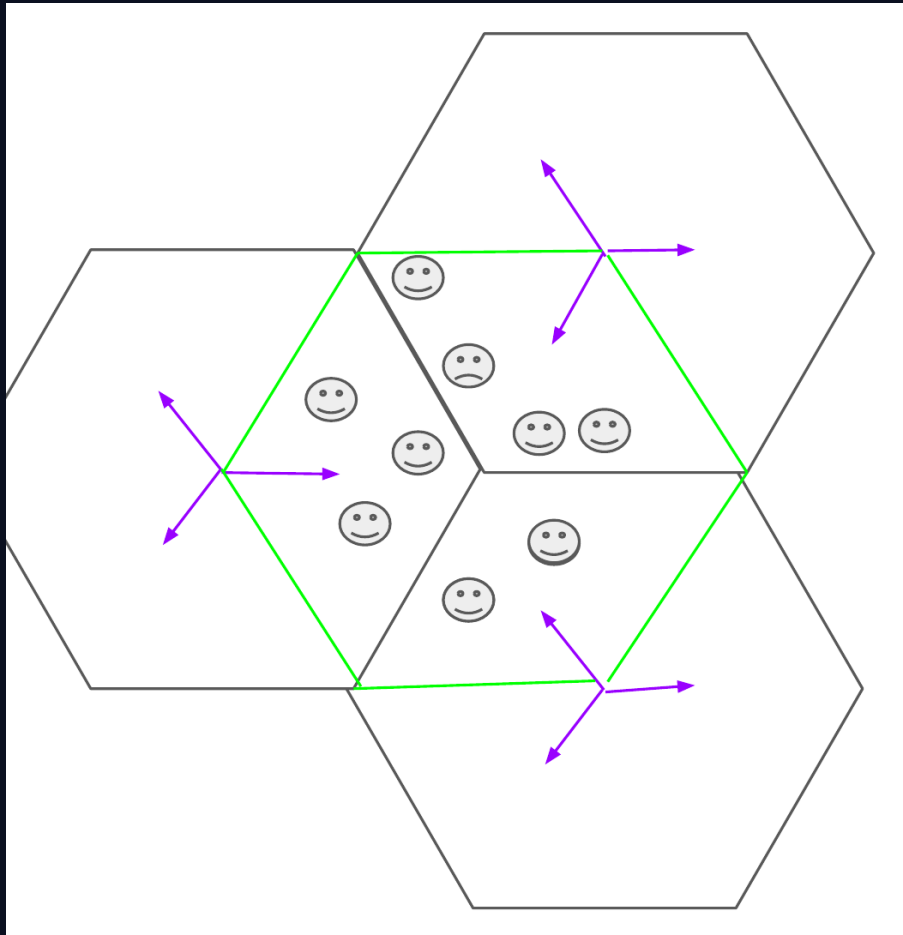
Tilting a city's antennas: k-means groups + a 12-qubit QAOA leaf

Real OpenCellID data → cost Hamiltonian → Classiq circuit → Wolfram exact simulation → Lean 4 proof

Team Quantum Headache

5-minute demo · September 2026

From a city of users to interaction regions and one utility U



Toy model: hexagonal cells, 120° sectors. The green hexagon is one interaction region R : three sectors from three sites that overlap.

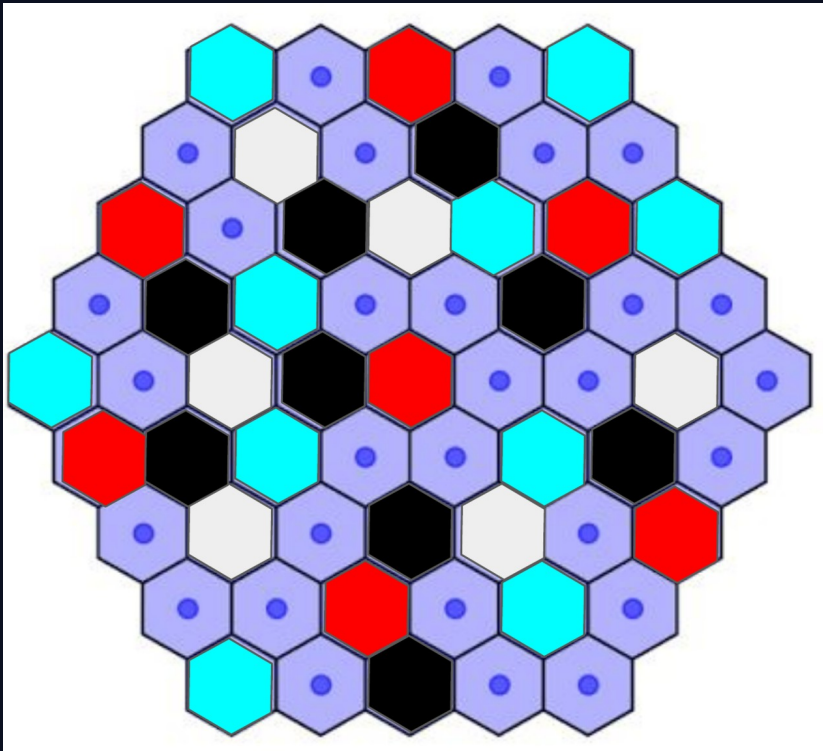
$$U(t) = \alpha \cdot Q(t) - \beta \cdot I(t) - \gamma \cdot D(t)$$

Q reception quality · I interference · D unmet demand (dead zones) — each summed over users, weighted by demand w_p

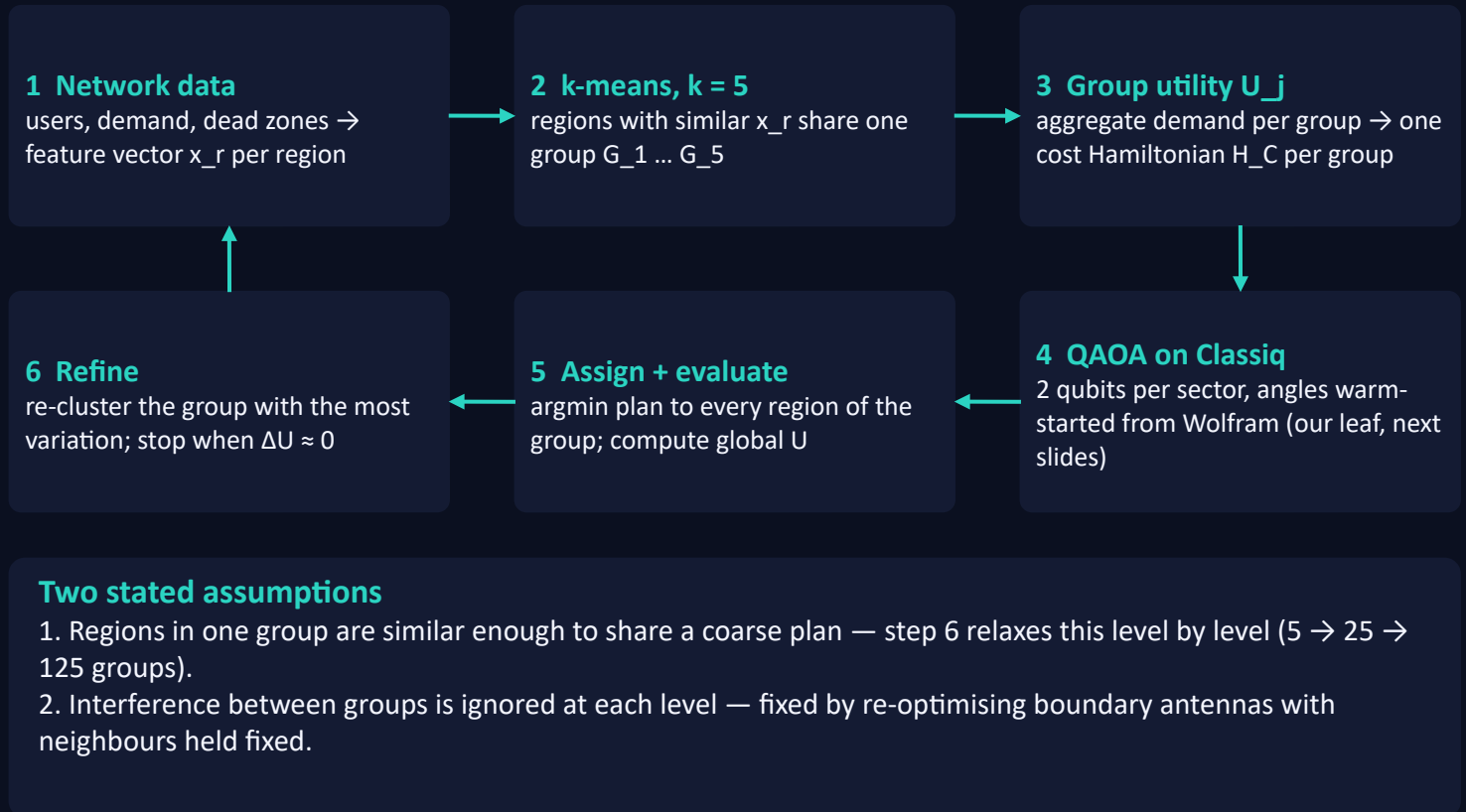
Our leaf cost is $-U$ with $\alpha = 1.5$ (coverage $c \cdot t$), $\beta = 1$ ($t^T I t$), $\gamma = 0.3$ (tilt penalty $\sum t$)

- ▶ The operator has spatial demand data: where the users are, where the dead zones are
- ▶ Each sector antenna gets one down-tilt; tilting down cuts interference on neighbours but shrinks coverage
- ▶ Every interaction region has its own optimum, and a city has thousands of regions
- ▶ Two ways to fail: one plan for the whole network (too coarse) or one job per region (too many jobs) → the hierarchy on the next slide

Whole network: k-means groups, one QAOA job per group

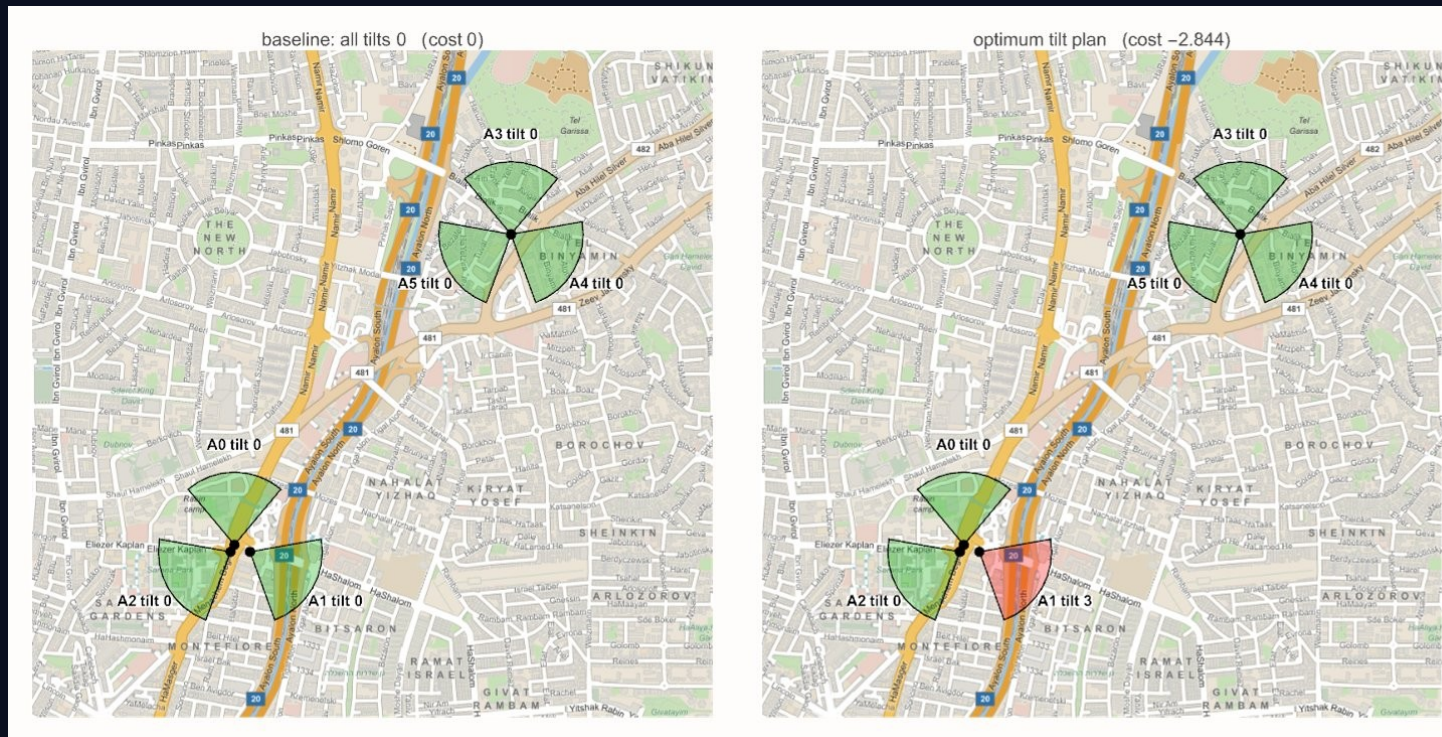


k-means ($k = 5$) on the toy network: 48 regions, 5 colours = 5 groups = 5 QAOA jobs.



Quantum work only where resolution is needed: ≤ 25 jobs per level = two batches of 15 parallel Classiq jobs. Classical: k-means + aggregation. Quantum: one warm-started QAOA per group.

One leaf of the tree: 6 real Tel Aviv antennas, one down-tilt each



Two real 3-sector sites from OpenCellID (Ayalon, Tel Aviv). Left: all tilts 0. Right: optimum plan — only A1 tilts to level 3.

- ▶ 6 sector antennas A0...A5, 4 tilt levels each $\rightarrow 4^6 = 4096$ plans
- ▶ Tilting down: less interference on neighbours, less coverage
- ▶ $\text{Cost}(t) = t^T | t - 1.5 \cdot c \cdot t + 0.3 \cdot \Sigma t$ (interference – coverage + penalty)
- ▶ Classical brute force: unique optimum -2.844 at $(0,3,0,0,0,0)$
- ▶ Real networks: 100s of antennas $\times$ 10 levels $\rightarrow$ intractable $\rightarrow$ quantum

Encoding: 2 qubits per antenna, 12 qubits total

$$\text{tilt}_i = q_{2i} + 2 \cdot q_{2i+1} \in \{0, 1, 2, 3\}$$

$q_i = (1 - Z_i)/2 \rightarrow$ Cost becomes a Pauli-Z polynomial

$$H_C = c_0 \cdot I + \sum h_i Z_i + \sum J_{ij} Z_i Z_j$$

73 terms: 1 identity + 12 Z + 60 ZZ — strictly 2-local, no ZZZ

- ▶ Why binary (2 qubits) and not one-hot (4 qubits) or a dense 5-qubit pack?
- ▶ Binary keeps H_C quadratic $\rightarrow$ only RZ and CX gates after ZZ decomposition
- ▶ Fits Classiq student edition comfortably; 15 parallel jobs used for sweeps
- ▶ QAOA layer: phase $\exp(+i\gamma H_C)$ then mixer $RX(2\beta)$ on every qubit
- ▶ Verified by Walsh–Hadamard: Pauli table reproduces all 4096 costs exactly

12

qubits

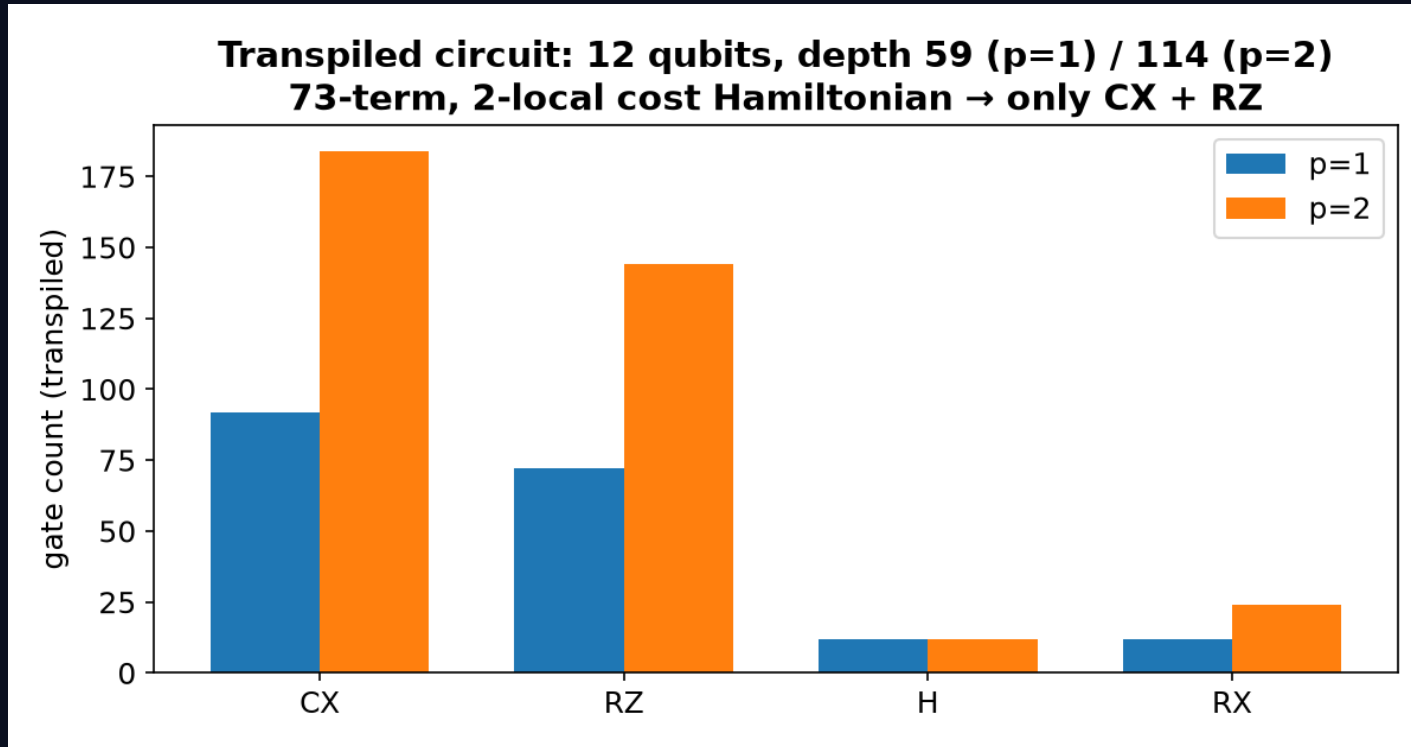
73

Pauli terms

2-local

ZZ max weight

The circuit Classiq synthesised

**59**depth, $p = 1$ **114**depth, $p = 2$ **92 CX**

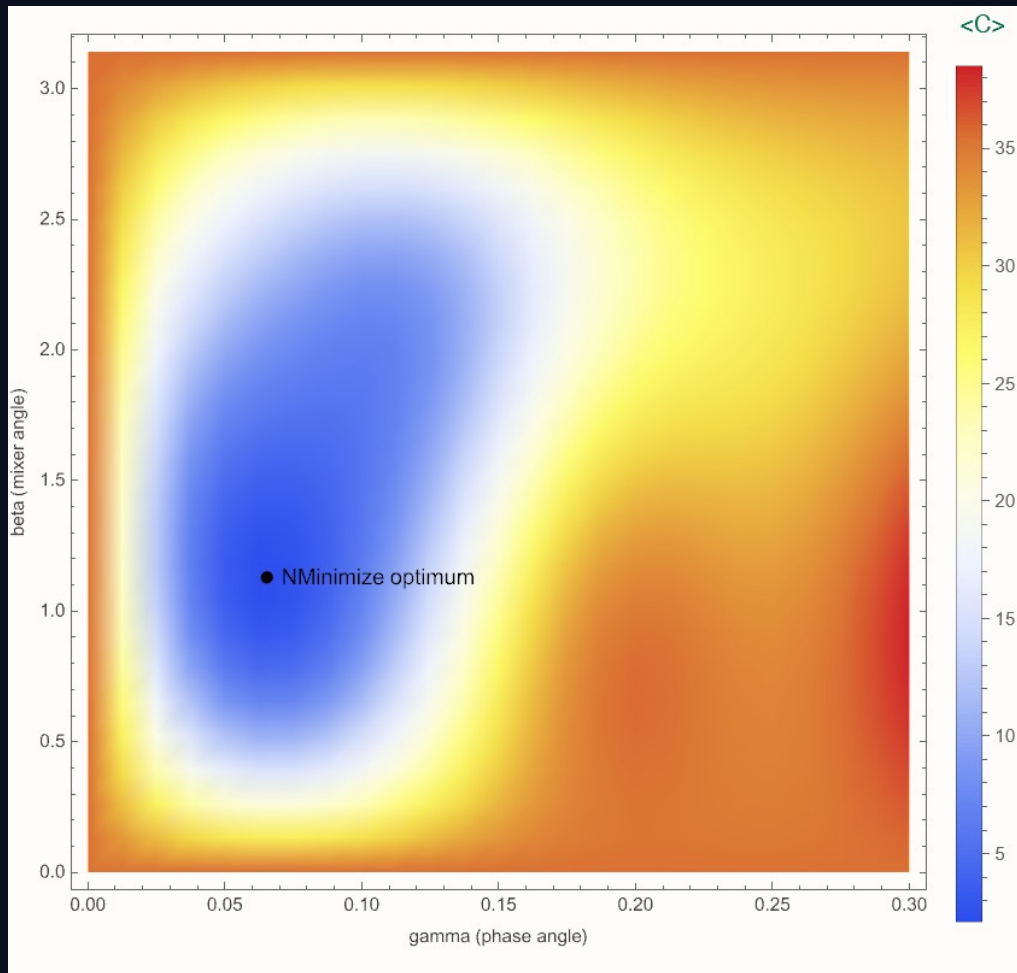
+ 72 RZ + 12 H + 12 RX

184 CX

+ 144 RZ + 12 H + 24 RX

- ▶ Written in Qmod: QStruct of QArray[QNum[2], 6], phase(cost, γ), RX mixer
- ▶ Exported as OpenQASM 3 (group/classiq/results/*.qasm) + hand-written QASM 2 layer
- ▶ Same circuit re-simulated in Qiskit statevector: $\langle C \rangle = 1.9985$ — identical to Wolfram

Wolfram exact simulation: where the good angles are



$(0.0654, 1.1293)$

optimal (γ, β) , $p = 1$

$\langle C \rangle = 1.9985$

$p = 1$ (random: 35.5)

$\langle C \rangle = -0.3825$

$p = 2, 4$ angles

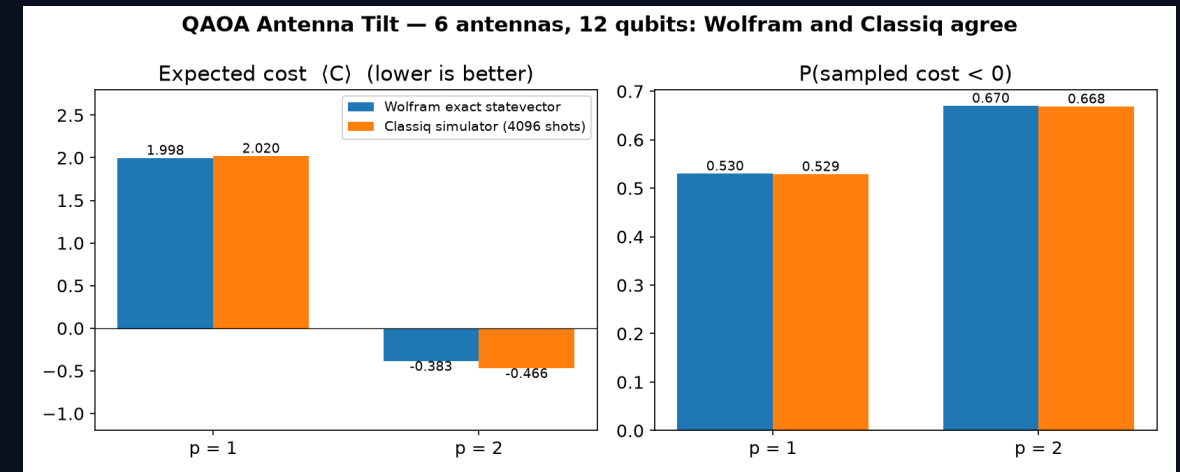
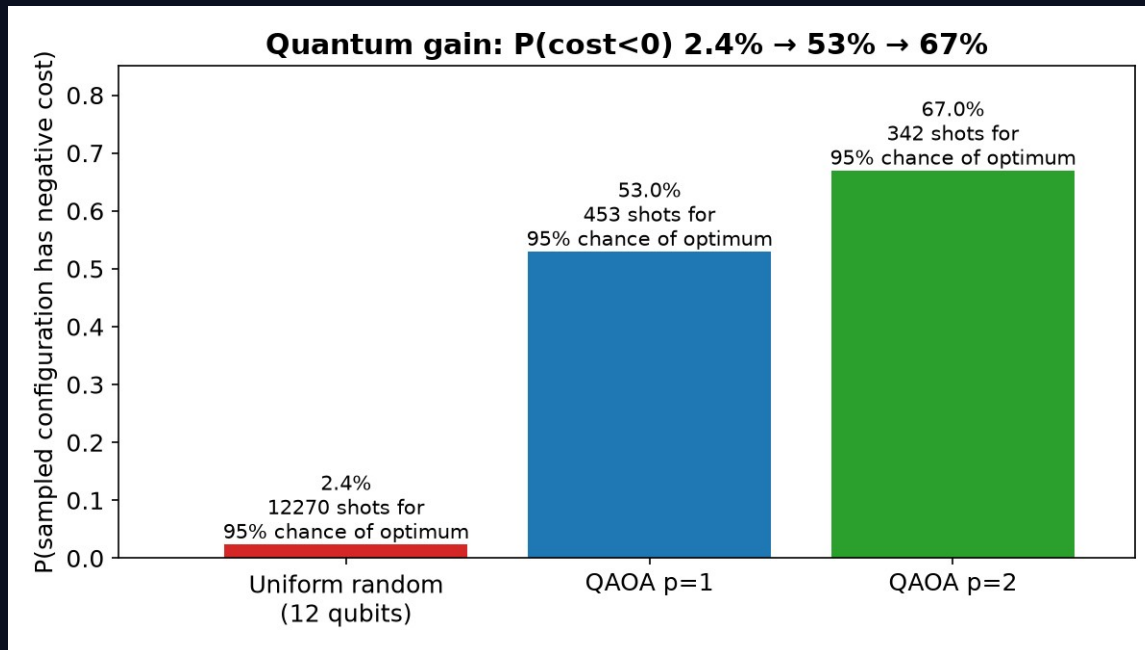
-2.844

ground state = classical optimum

- ▶ Notebook + PDF + CDF + Markdown in [group/wolfram/](#) (14 pages, 13 figures)
- ▶ Landscape is flat almost everywhere — a bad start sits on a plateau at $\langle C \rangle \approx 35$
- ▶ Angles found here are the warm start we hand to Classiq
- ▶ Also in the notebook: VQE (RY+CZ), Grover on a 2-antenna slice, 10-level scaling

$\langle C \rangle(\gamma, \beta)$ for $p = 1$ over all 4096 amplitudes (statevector, no shot noise).

What the quantum sampler buys us, and Classiq reproduces it



- Best sampled bitstring in every Classiq run = the classical optimum (0,3,0,0,0,0)
- $p = 2$: $\langle C \rangle = -0.466$ sampled vs -0.383 exact; 716 distinct plans in 4096 shots

2.4% $\rightarrow$ 67%

$P(\text{cost} < 0)$: random $\rightarrow$ $p = 2$

12270 $\rightarrow$ 342

shots for 95% chance of the optimum

2.020 vs 1.999

$\langle C \rangle$ $p = 1$: Classiq 4096 shots vs exact

0.529 vs 0.530

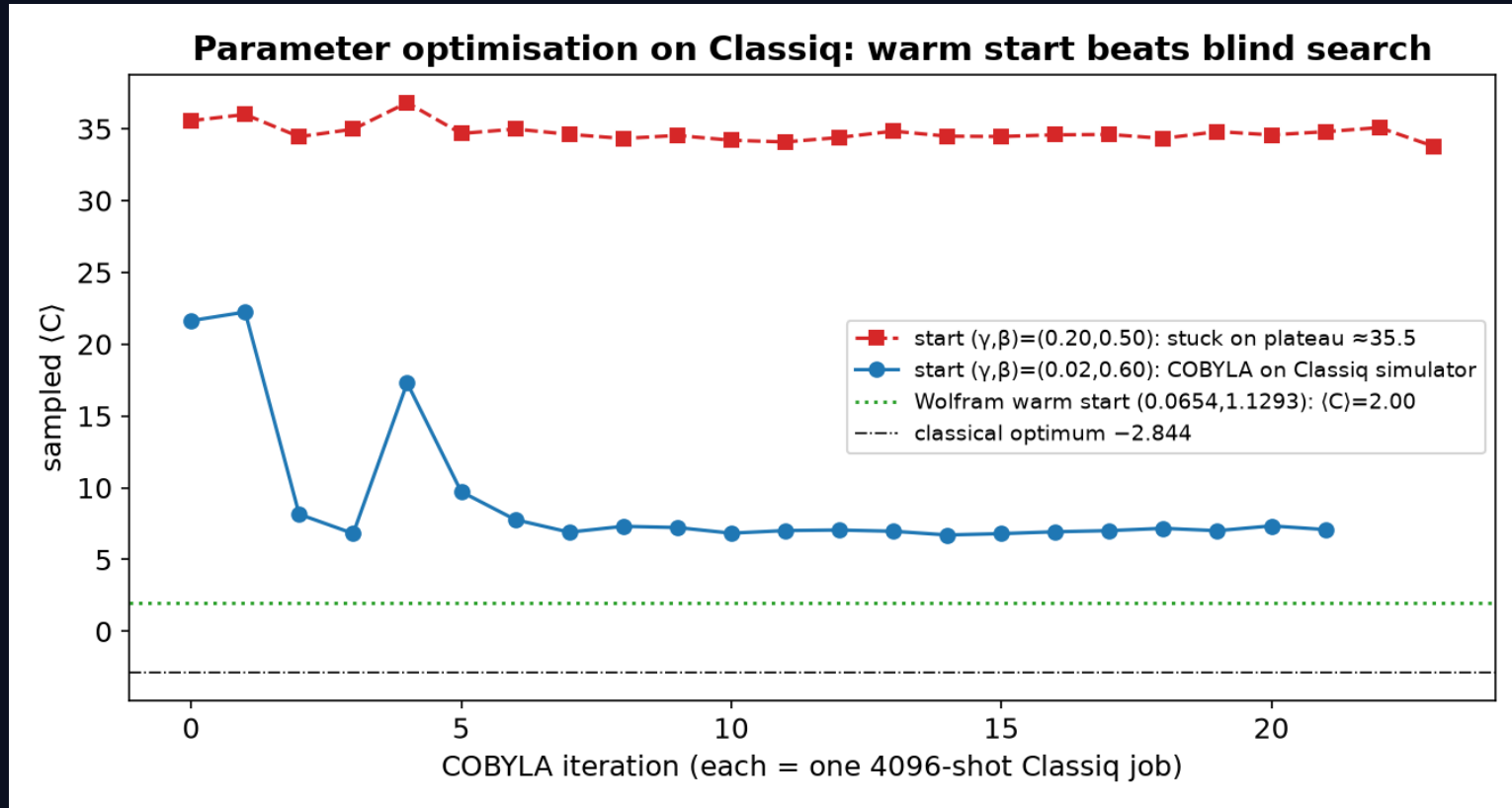
$P(\text{cost} < 0)$ $p = 1$: Classiq vs exact

Demo: the QAOA distribution sharpening



- ▶ 120 frames, angles ramp from 0 to the $p = 2$ optimum
- ▶ Scatter: probability of each of the 4096 plans vs its cost
- ▶ Bars: $P(\text{cost} < 0)$ climbing to 0.67
- ▶ Black line: classical optimum -2.844
- ▶ Second clip:
group/video/classiq_convergence.mp4 —
COBYLA on Classiq, iteration by iteration

Optimising on Classiq: warm start beats blind search



- ▶ Start $(0.20, 0.50)$: COBYLA stuck at $\langle C \rangle \approx 35.5$ — the plateau Wolfram predicted
- ▶ Start $(0.02, 0.60)$: 20 Classiq jobs, $\langle C \rangle 21.6 \rightarrow 7.0$, still above warm start 2.0
- ▶ Shot noise (4096 shots) hides gradients; 15 parallel jobs let us sweep starts instead
- ▶ Takeaway: exact classical pre-simulation of small instances gives angles that transfer

Click to play: 20 COBYLA iterations on the Classiq simulator, one 4096-shot job each (11 s).

Lean 4 certificates: the classical facts are machine-checked

```

theorem decodeTilt_bijective : Function.Bijective (fun q : Fin 2 × Fin 2 => decodeTilt q.1 q.2) -- 2 qubits ↔ 4 tilts
theorem costS_optTilt       : costS optTilt = -28440 -- cost ×104 (decide)
theorem costS_lower_bound   : ∀ t : Fin 6 → Fin 4, -28440 ≤ costS t -- 4096 cases (native_decide)
theorem costS_unique        : ∀ t : Fin 6 → Fin 4, costS t = -28440 → t = optTilt -- optimum is unique
theorem cost_phase_preserves_probability : ∀ γ c a, ‖exp(γ c i) · a‖2 = ‖a‖2 -- phase layer keeps P
theorem two_smul_P0 / two_smul_P1 : 2•P0 = 1 + Z, 2•P1 = 1 - Z -- qubit → Pauli-Z

```

0 sorry

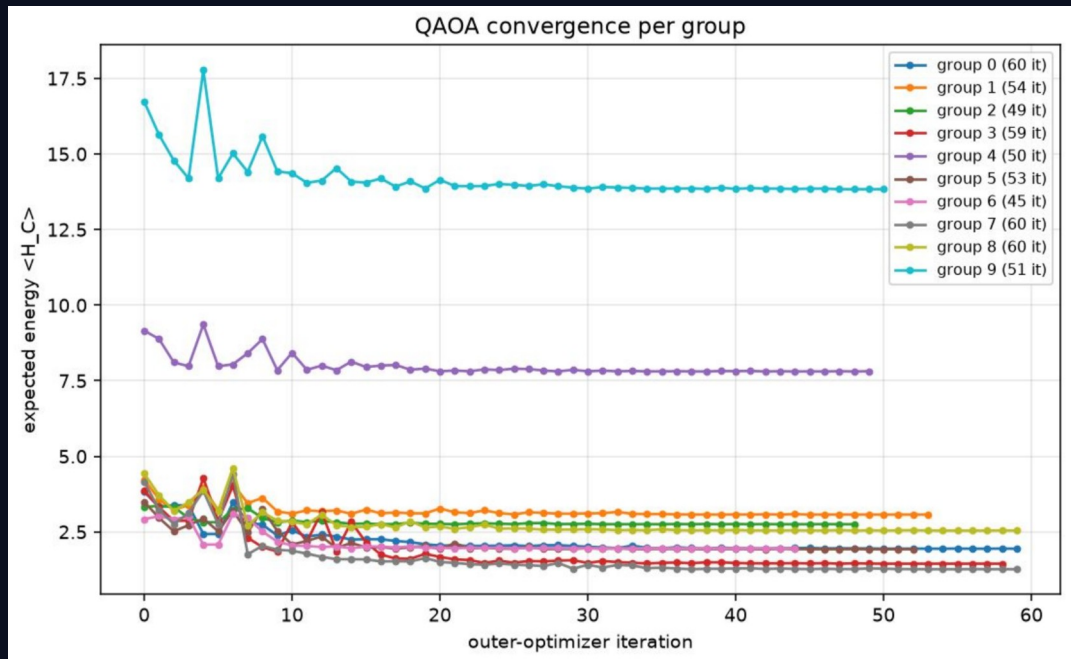
lake build succeeded

Mathlib

Lean 4.33.1

- ▶ group/lean/TiltQAOA.lean + build log
- ▶ Proves the optimum, its uniqueness and the encoding are exactly what the quantum runs report
- ▶ Quantum layer: unitarity of the phase operator, so QAOA never destroys probability mass

Why 2 qubits per sector, not a 5-qubit pack of 27 configurations



Group run, 5-qubit encoding (27 of 32 states valid): $\langle H_C \rangle$ converges in all 10 groups, but the sampled optimum is no more likely than uniform.

The hierarchy is encoding-agnostic: keep k-means + refinement, swap the leaf solver. The group's own next step ("6-qubit encoding, constrained mixer") is this binary encoding — already run on Classiq and proven in Lean.

0.5 – 1.2×

$P(\text{optimum}) / \text{uniform, 5-qubit pack, } p = 1, 3, 6$

27×

$P(\text{optimum}) / \text{uniform, 2 qubits per sector, } p = 1$

15.6% → 3.0%

invalid states, 5-qubit pack, uniform → $p = 3$

0%

invalid states, binary: every bitstring is a plan

- ▶ Dense pack: 5 penalised states out of 32, so the QAOA layers go into learning the penalty, not the utility
- ▶ Circuit cost is not the issue: depth 56 on 5 qubits vs 59 on 12 qubits, RZ + CX in both
- ▶ Binary per sector keeps H_C 2-local; $P(\text{cost} < 0)$ climbs 53% → 67% from $p = 1$ to 2
- ▶ And it scales: 10 tilt levels = 4 qubits per sector, 24 qubits for 6 sectors, gate count grows quadratically

Why quantum: what changes at 1000 qubits (500 antennas)

$$2^{1000} \approx 10^{301}$$

plans for 500 antennas $\times$ 4 tilts (today: 4096)

$$\approx 45 \text{ qubits}$$

ceiling of exact statevector simulation: 50 q = 18 PB > Frontier 9 PB

$$\approx 13\,000 \text{ CX}$$

one QAOA layer at 1000 q: 2-local, local coupling, depth $\sim 10^2$

Best non-quantum tools on this size

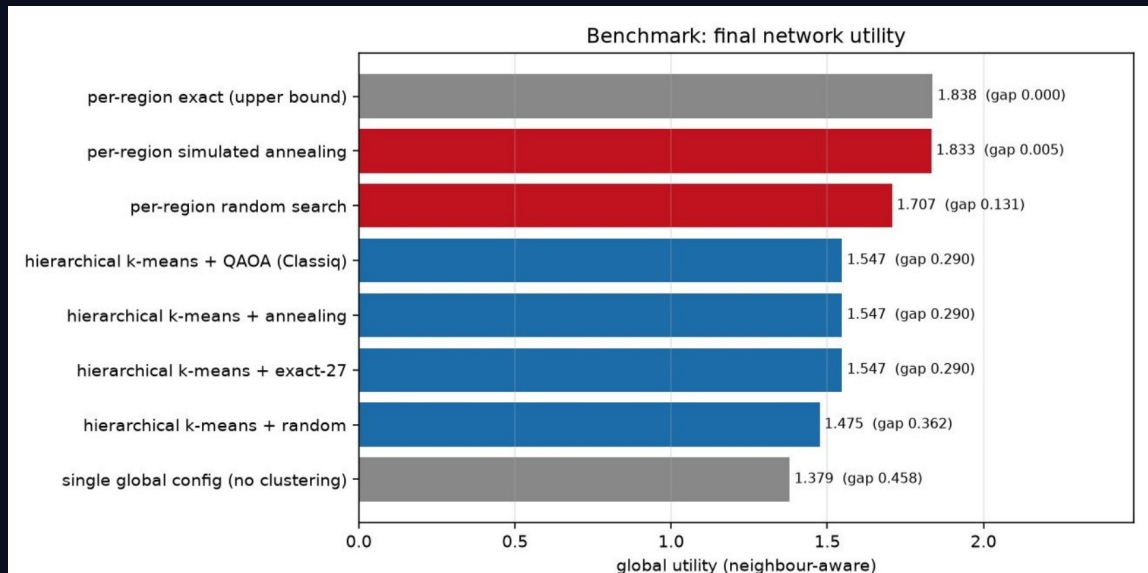
- ▶ Brute force (our exact check today): 10^{301} plans at 10^{18} / s (exascale) = 10^{283} s; the universe is 4×10^{17} s old
- ▶ Exact simulation of the quantum algorithm (Wolfram / Classiq simulator): memory $2^n \times 16$ B, 45 q = 0.5 PB, 50 q = 18 PB. Nobody can simulate QAOA past ≈ 48 qubits
- ▶ Exact QUBO solvers (branch & bound, Gurobi): sparse 1000-variable instances run hours to days and may never certify the optimum
- ▶ Heuristics (simulated annealing, tabu, D-Wave 4 400-qubit annealer): minutes, good plan, no guarantee. This is the real competitor (1.833 vs 1.838 on the toy benchmark)

A real 1000-qubit gate-based quantum computer

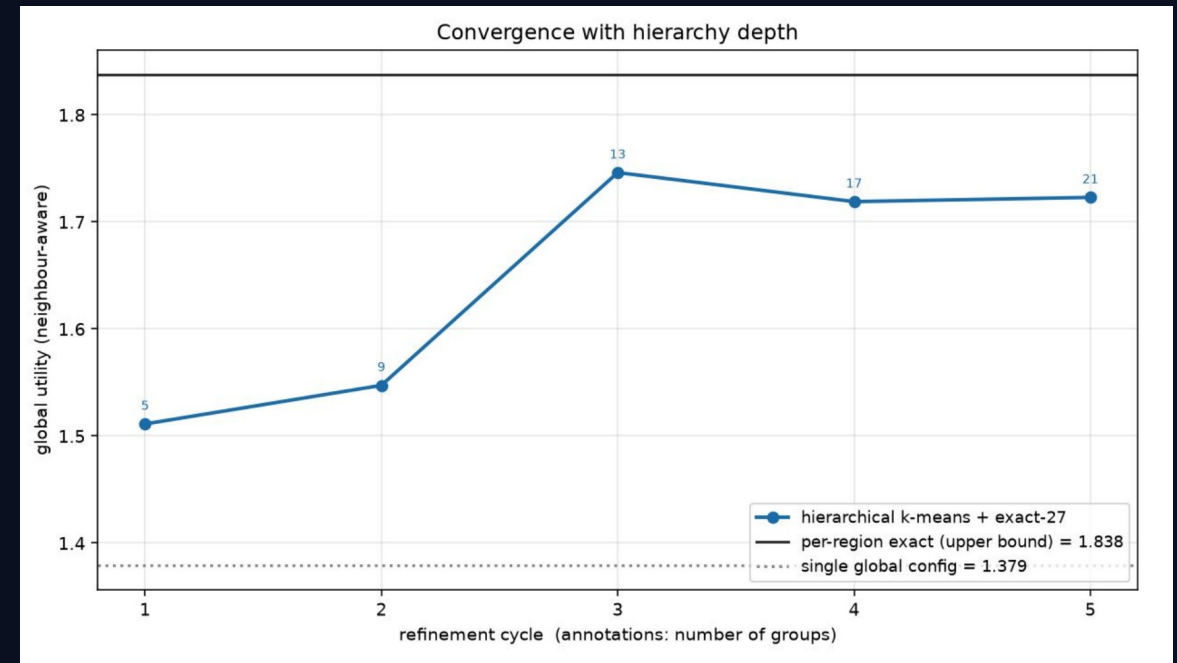
- ▶ Physical qubits exist (Sep 2026): IBM Condor 1 121, Atom Computing 1 200, Infleqtion 1 600; best 2-qubit fidelity 99.92% (Quantinuum Helios, 98 q)
- ▶ 13 000 CX at 99.9% = ≈ 13 errors per shot, so the run must be error-corrected: 1000 logical qubits = 2 000 (2:1 colour code) to 100 000 (surface code) physical. Record today: 96 logical (QuEra). Roadmaps: 2028–2030
- ▶ Runtime once available: depth $\sim 10^2$ per layer, 1 shot ≈ 0.1 ms (superconducting) to 10 ms (ions); 10^4 shots $\times$ 20 iterations = seconds to minutes; Wolfram warm start cuts the iterations
- ▶ Same Qmod program: Classiq synthesises for the target, simulator today, hardware backend tomorrow. No code change but the backend

Classical tools give a good plan and stop there. At 12 qubits we measured 27 $\times$ amplification and $P(\text{cost} < 0)$ 2.4% $\rightarrow$ 53% $\rightarrow$ 67% with p; whether that persists at 1000 qubits cannot be checked classically. The only instrument that can run the algorithm at this size is the quantum computer itself.

Benchmark: 84% of the ceiling with 4.8× fewer quantum jobs



Hierarchy + QAOA equals exact-27 and annealing on every group (10 / 10 optimal).



Refinement 5 → 21 groups; cycle 3 (13 groups) already reaches 1.75 = 95% of the ceiling.

1.379

one config for all · 1 job

1.547

hierarchy + QAOA · 10 jobs

1.838

per-region ceiling · 48 jobs

Video: youtu.be/OSetTWnRQhc

Code:

github.com/zuwasi/quantum-headache-antenna-tilt

Team Quantum Headache — thank you. Questions?